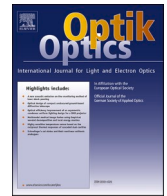




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Original research article

High precision indoor positioning algorithm of single LED lamp based on A-Bayes

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ABSTRACT

With the rapid development of information technology, the application field of city positioning technology is more and more extensive. Traditional indoor positioning technology has shortcomings in positioning accuracy, equipment cost, security and so on. Visible light indoor positioning technology can effectively solve the above problems. This paper presents an Adaptive boosting- Bayes (A-Bayes) localization algorithm. In this algorithm, three tilted photodetectors and one horizontal PD are used as receivers to measure the received optical power of each PD and set up a fingerprint database. The Adaboost strong classifier is used to divide the test space into two regions: the center and the edge. In the online positioning stage, for the measured received optical power, the region to which it belongs is first determined, and then the Bayesian matching algorithm is used to match the regions to complete the positioning. The experimental results show that in the test space of, and the overall average positioning error is 9.27 cm. Compared with the traditional indoor positioning algorithm, the positioning error of the proposed algorithm is smaller; Compared with the traditional multi lamp location algorithm, the algorithm proposed in this paper avoids the interference between multi light source symbols, and the equipment is simpler; Compared with the traditional fingerprint matching algorithm, the proposed algorithm has faster positioning speed.

1. Introduction

In recent years, global positioning system (GPS) is widely used in vehicle navigation, map building and other fields. But the complex indoor environment will lead to GPS positioning error is very large [1–3]. With the development of science and technology, many indoor positioning technologies are applied, such as infrared, WiFi (wireless fidelity), radio frequency identification (RFID), Bluetooth, ultra wide band (UWB), etc. [4–6]. These traditional radio frequency positioning technologies generally have the disadvantages of low positioning accuracy and expensive equipment. Using visible light for positioning is a new idea to solve the problems of traditional indoor positioning technology [7–10]. Compared with RF signal, visible light has better security, wide frequency band and high positioning accuracy, which has become a research hotspot [11–13]. Fingerprint location algorithm is popular because of its high location accuracy, simple algorithm and low equipment price. Fingerprint location algorithm is mainly divided into two kinds: one is the nearest neighbor method, the other is Bayesian matching algorithm [14]. Compared with the nearest neighbor method, Bayesian matching algorithm has higher positioning accuracy. However, the two matching algorithms have common shortcomings. They need to match each fingerprint in the fingerprint database established in the offline stage. When the test area is large, the positioning time

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will be very long [15]. So how to speed up the positioning speed is a problem to be considered at this stage.

In recent years, many researchers combine machine learning algorithm with visible light indoor positioning, and obtain good positioning results: In 2019, Sheng Zhang et al. Proposed a three-dimensional indoor positioning system based on RSS PDOA hybrid artificial neural network. Simulation results show that ann-pdoa positioning system has better robustness, and the positioning error of 80% of test points is within 8 cm [16]. In 2019, Chuang, et al. proposed a positioning algorithm that uses a combination of repeating units and machine learning. They discussed the second-order regression ML algorithm and the polynomial three-sided ML algorithm respectively, and achieved centimeter-level positioning accuracy [17]. In 2019, Zhang et al. proposed a high-precision indoor positioning algorithm based on sparse training Bayesian regularization points, and the positioning error can reach about 4 cm [18]. In 2020, Mao, W et al. proposed a high-precision indoor visible light positioning system based on an improved Camshift tracking algorithm. The positioning error can reach about 2 cm, but the algorithm involves image processing and is more complicated [19]. It can be seen from the above research status that most of the current positioning algorithms combine machine learning with multiple LED lights. Although good positioning accuracy is achieved, it needs to distinguish light sources, and the equipment is more complex [20].

Aiming at the slow positioning speed of traditional fingerprint matching algorithm and the problems of using machine learning algorithm combined with multiple lights for positioning, this paper uses A-Bayes algorithm combined with a single LED light to achieve high-precision and fast positioning in a large area. In this algorithm, the receiver consists of one horizontal PD and three tilted PD. In the off-line phase, the received optical power of each PD is measured under the direct line of sight link and the reflected line of sight link to eliminate the weak low reflection signal and establish the fingerprint database. AdaBoost strong classifier is used to divide the test space into center and edge regions, and fingerprint databases belonging to edge and center are established respectively. In the online positioning stage, for the measured received optical power, AdaBoost strong classifier is used to determine the region it belongs to, and then Bayesian algorithm is used to match the regions to complete the positioning.

This article is arranged as follows: the second part introduces the principle of the positioning algorithm, the third part introduces the algorithm simulation results, the fourth part introduces the experimental test results, and the fifth part summarizes the article as a whole.

2. Principle of location algorithm

The indoor positioning model is shown in my paper [13]. In paper [13], only direct line of sight (LOS) links are considered. On this basis, two kinds of Los links, direct line of sight (LOS) and reflection line of sight (LOS), are considered.

Under the reflected line-of-sight link, the channel gain can be expressed as:

$$H_{NLOS} = \begin{cases} \frac{(m+1)A}{2\pi l_1^2 l_2^2} \rho \cdot \cos^m \varphi \cdot dA_{ref} \cos(\lambda) \cos(\gamma) \cos(\psi_r) g(\psi_r) & 0 \leq \psi_r \leq FOV \\ 0 & \text{otherwise} \end{cases} \quad (1)$$

In Eq. (1), l_1 is the distance from the LED light to the reflecting surface element, l_2 is the distance from the reflective surface element to the PD. ρ is the reflection coefficient of the wall, dA_{ref} is the area of the reflecting surface element, λ is the receiving angle of the reflecting surface element, and is the emitting angle of the reflecting surface element.

According to $P_{ri} = P_t \cdot H$, the total received optical power can be expressed as:

$$P_{ri} = P_t \cdot H_{LOSi} + P_t \cdot H_{NLOSi} \quad (2)$$

In Eq. (2), P_t is the emission power of the LED light source, which is a known quantity. Assuming there are N test points, the average value of the received optical power can be expressed as:

$$\bar{P}_{ri} = \frac{1}{N} \sum_{i=1}^N P_{ri} \quad (3)$$

To eliminate low reflection signals, the selected received optical power satisfies the following relationship:

$$P_{ri} \geq (1 + \alpha) \cdot \bar{P}_{ri} \quad (4)$$

In Eq. (4), α is the outlier ratio. In this way, four received optical power values will be obtained at each test point: $P_{r1}, P_{r2}, P_{r3}, P_{r4}$. Then the light intensity fingerprint library established by N test points is:

$$P = \begin{pmatrix} P_{r11} & P_{r12} & P_{r13} & P_{r14} \\ P_{r21} & P_{r22} & P_{r23} & P_{r24} \\ \dots & \dots & \dots & \dots \\ P_{rN1} & P_{rN2} & P_{rN3} & P_{rN4} \end{pmatrix} \quad (5)$$

After obtaining the fingerprint database, select L positions as the training set, and the rest as the test set, then the training set is:

$$P_{train} = \begin{pmatrix} P_{r11} & P_{r22} & P_{r13} & P_{r14} \\ P_{r21} & P_{r22} & P_{r23} & P_{r24} \\ \dots & \dots & \dots & \dots \\ P_{rL1} & P_{rL2} & P_{rL3} & P_{rL4} \end{pmatrix} \quad (6)$$

The test set is:

$$P_{test} = \begin{pmatrix} P_{rL+11} & P_{rL+12} & P_{rL+13} & P_{rL+14} \\ \dots & \dots & \dots & \dots \\ \dots & \dots & \dots & \dots \\ P_{rN1} & P_{rN2} & P_{rN3} & P_{rN4} \end{pmatrix} \quad (7)$$

After obtaining the training set and the test set, the light intensity fingerprint data in the training set is divided into two categories: edge and center through the Adaboost algorithm. The specific area division process is as follows:

- (1) Use the light intensity fingerprint data in the training set as the initial sample set D.
- (2) Select base learner: In this article, select SVM classifier as the base learner.
- (3) Initialize the number of training rounds and set the number of training rounds to T.
- (4) Initialize the sample weight distribution:

$$D_1(x) = \frac{1}{N}$$

- (5) Train the classifier from D, the classifier is represented as $h_t = \zeta(D, D_t)$.
- (6) Estimate the probability of classification error: $\epsilon_t = P_{x \sim D_t}(h_t(x) \neq f(x))$.
- (7) Determine the weight of the classifier: $\alpha_t = \frac{1}{2} \ln \left(\frac{1-\epsilon_t}{\epsilon_t} \right)$
- (8) Update the sample distribution, Z_t is the normalization factor, and the expression of the sample distribution is:

$$D_{t+1}(x) = \frac{D_t(x)}{Z_t} \times \begin{cases} \exp(-\alpha_t), & h_t(x) = f(x) \\ \exp(\alpha_t), & h_t(x) \neq f(x) \end{cases}$$

- (9) After updating the sample distribution, repeat the above steps until the number of training rounds is reached.

Through the above Adaboost algorithm, the sample set can be divided into edge and center categories. The reason for dividing the sample set into two categories is to speed up the positioning speed. According to the location of the region divided by the Adaboost strong classifier, the fingerprint database of the internal region and the external region are respectively established for A-bayes sub-region fingerprint matching.

The localization algorithm adopts Bayesian matching algorithm. First, input the test set into the trained Adaboost classifier to determine whether the input data belongs to the edge area or the central area, and then perform Bayesian analysis on the edge or central area with the established internal and external fingerprint libraries. Si matching avoids the problem of slow positioning speed caused by global matching. The specific positioning process is as follows:

Each test location has a prior probability $P(l)$, which is equal at each test point. When the fingerprint l_f is known, the conditional probability of the location is as follows:

$$P(l|l_f) = \frac{P(l_f|l)P(l)}{\sum_{l_k \in L_{oc}} P(l_f|l_k)P(l_k)} \quad (8)$$

We use the position information with the largest posterior probability as the estimated position:

$$L = \arg\max_{l_k \in L_{oc}} P(l_k|l_f) = \arg\max_{l_k \in L_{oc}} P(l_f|l_k)P(l_k) \quad (9)$$

Since the received optical power satisfies the Gaussian distribution, four received optical powers can be measured at each grid point, and the received optical powers between different PDs are independent of each other, so the joint probability distribution function $P(l_f|l_k)$ can be expressed as:

$$P(l_f|l_k) = P(l_{f1}|l_k)P(l_{f2}|l_k)P(l_{f3}|l_k)P(l_{f4}|l_k) \quad (10)$$

In Eq. (10), $P(l_{f1}|l_k)$, $P(l_{f2}|l_k)$, $P(l_{f3}|l_k)$ and $P(l_{f4}|l_k)$ respectively are the received optical power of PD_i (i = 1,2,3,4) at a certain grid point. Putting Eq. (10) into Eq. (9), you can Calculate the maximum posterior probability.

The positioning coordinates (x_r, y_r) are obtained through the Bayesian matching algorithm, and the final positioning error is calculated as:

$$error = \sqrt{(x - x_r)^2 + (y - y_r)^2} \quad (11)$$

3. Positioning simulation system

This paper constructs a simulation system to evaluate the performance of the positioning system, as shown in Fig. 1. The length, width, and height of the initial positioning area are set to 3 m, 3 m, and 3 m respectively, and the coordinates of the LED light source are (1.5, 1.5, 3). The initial value of the distance r between the center of the horizontal PD and the center of the inclined PD is set to 1 cm, the azimuth difference θ between two adjacent inclined PDs is set to 90° , and the initial values of the placement angles ω_1 , ω_2 and ω_3 of the three inclined PDs are respectively 0° , 90° , 180° . The area to be tested is divided into 361 grids of equal size. In the offline phase, the intersection of each grid is used as a test point to test the received optical power and establish a fingerprint database. In the online positioning stage, the position 2 cm away from each intersection is used as the test point. This article considers the reflection at the far right of the room, and the reflection coefficient is set to 0.1.

Under the direct line-of-sight link and the reflected line-of-sight link, the contour map of the total received optical power is shown in Fig. 2 below:

It can be seen from Fig. 2 that when the reflection on the right side of the room is considered, the total power distribution of the received light of the four PDs under the direct line-of-sight link and the reflected line-of-sight link meets the strength in the middle and the weak in the surroundings, and the central area is directed toward the reflective surface. The law of one side offset.

When the total received optical power under the direct line-of-sight link and the reflected line-of-sight link is calculated, the establishment of the light intensity fingerprint library is completed, and the Adaboost strong classifier is used to divide the room test area. The results of the division are shown in Fig. 3:

It can be seen from Fig. 3 that the region division result of Adaboost strong classifier is consistent with the regional rule shown in the medium high plane map in Fig. 2, and the central region shifts to the side of the reflector. Simulation results show that the classification accuracy of AdaBoost strong classifier is 98.89%.

In this paper, the test set data is located by a-bayes zonal matching algorithm, and the positioning time of global nearest neighbor method, regional nearest neighbor matching method, global Bayesian matching method and regional Bayesian matching method in the test points of 10, 20, 30, 40, 50, 60 are compared, as shown in Fig. 4 below.

In Fig. 4, the box, circular, upper trigonometric and lower trigonometric curve represent the global nearest neighbor matching method, the regional nearest neighbor matching method, the global Bayesian matching method, and the regional Bayesian matching method respectively. It can be seen that when the number of test points is the same, the location speed of A-bayes zonal matching algorithm is the fastest and that of global nearest neighbor matching is the slowest. With the increase of the number of test points, the location time of the four location algorithms is increasing. Among them, the location time of A-bayes partition matching algorithm is the slowest, and the location time of global nearest neighbor matching algorithm is the fastest. Therefore, when the number of test points is large, using A-bayes partition matching algorithm can significantly shorten the location time, and the effect is better.

This paper also uses the nearest neighbor method and Bayesian matching algorithm to compare the average positioning error of the first 60 test points as a whole and compare them by region, as shown in Table 1:

It can be seen from Table 1 that when the first 60 test points are tested, the positioning error is smaller by using the A-Bayes partition matching algorithm proposed in this paper, which indicates that the positioning algorithm proposed in this paper can not only speed up the positioning speed, but also reduce the positioning error, and the positioning effect is better than the traditional fingerprint matching algorithm.

When all grid points in the area to be tested are tested, the location error distribution map and error histogram of A-bayes zonal matching algorithm are shown in Fig. 5(a) and (b), respectively.

It can be seen from Fig. 5(a) that the maximum positioning error of the A-bayes sub-region matching algorithm does not exceed

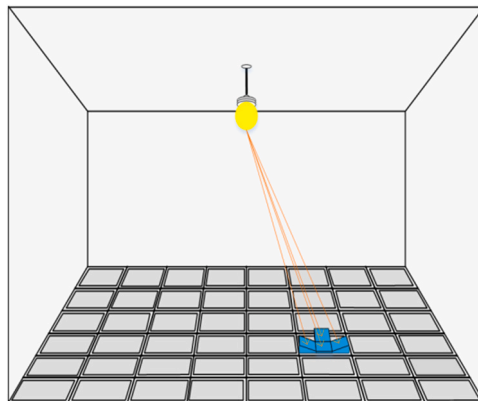


Fig. 1. Positioning system model.

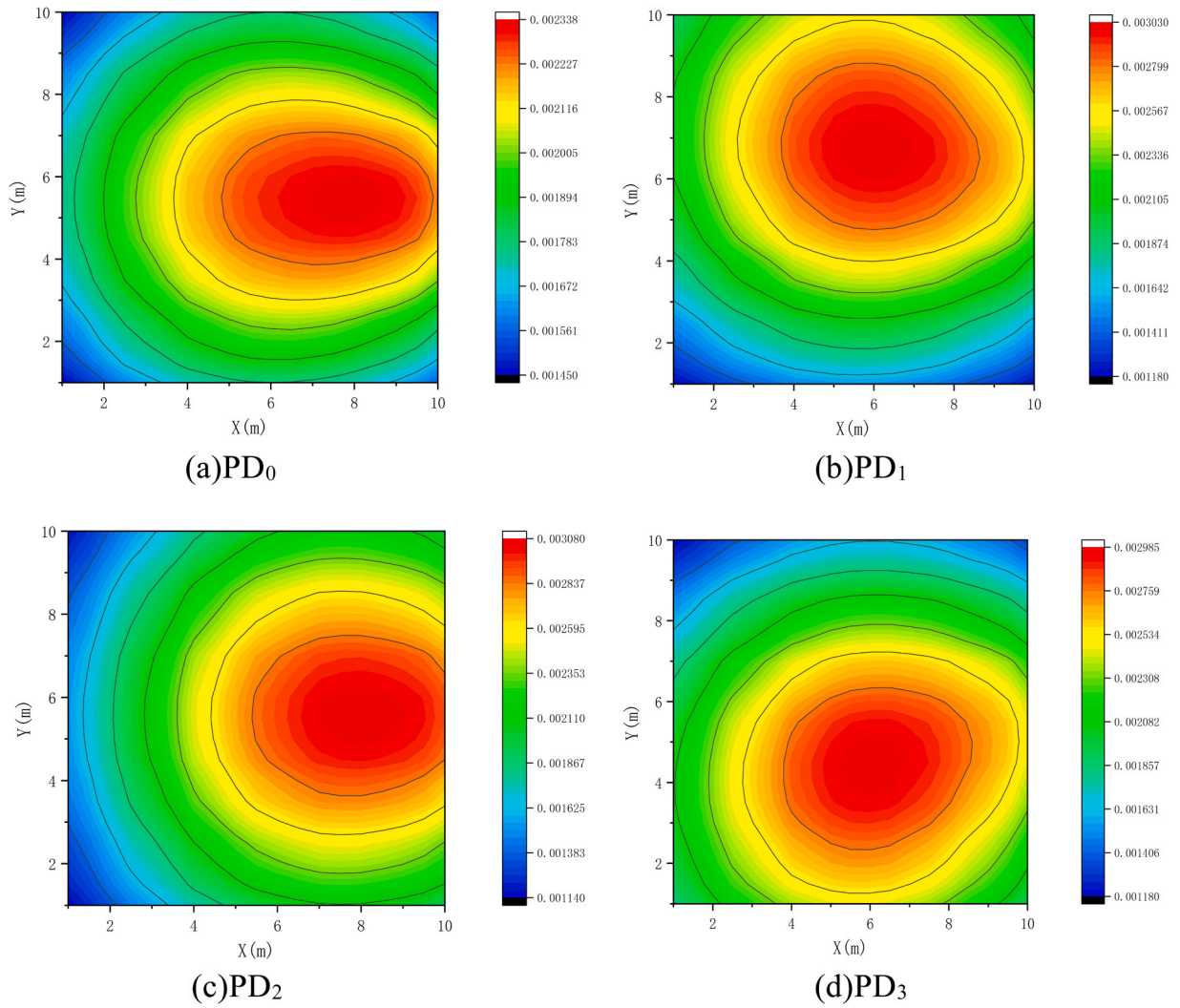


Fig. 2. Contour surface distribution of the received optical power of four PDs.

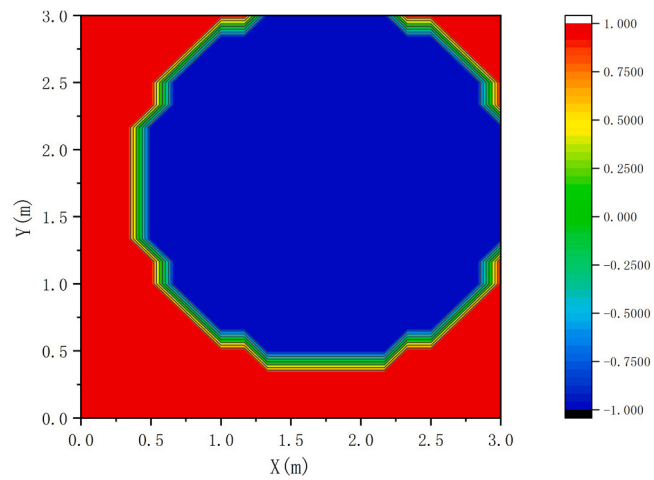


Fig. 3. partition results of AdaBoost strong classifier.

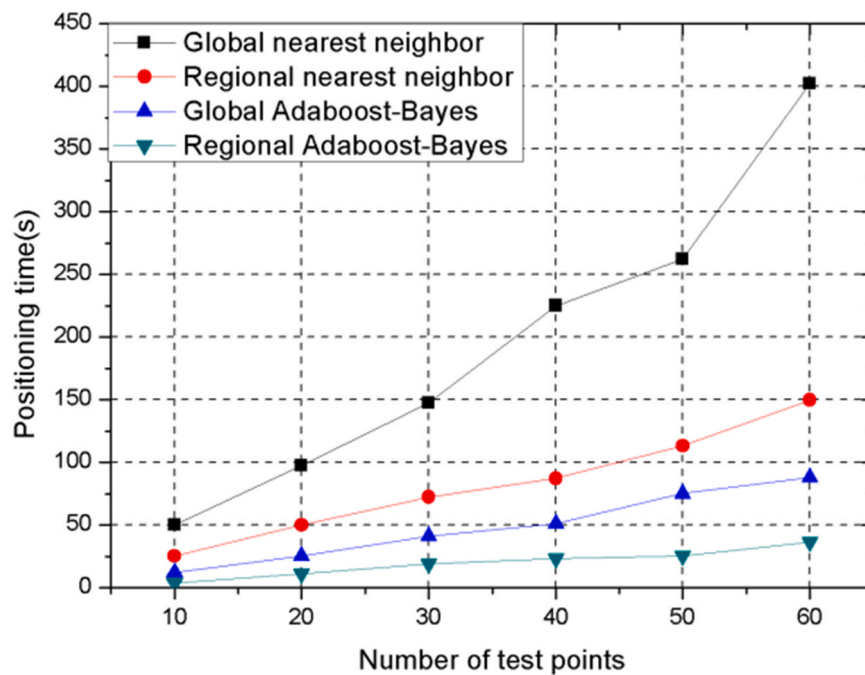
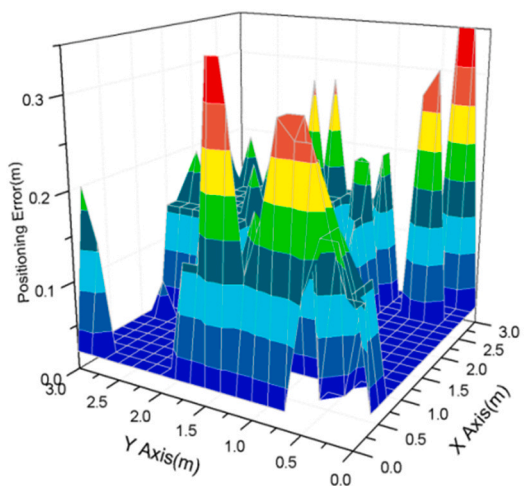


Fig. 4. Comparison of positioning time of different matching methods. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

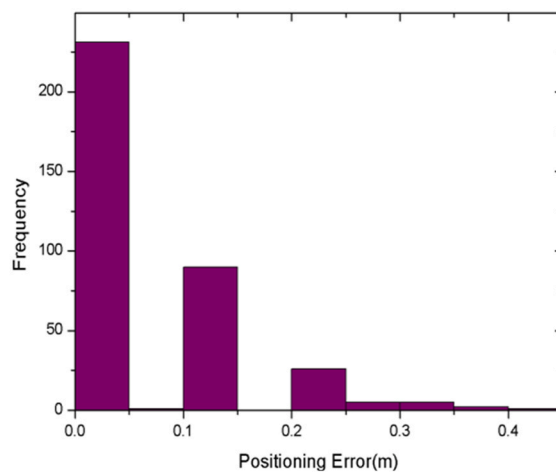
Table 1

Comparison of the average positioning errors of the four matching algorithms.

Algorithm category	Global nearest neighbor method	Global Bayesian matching method	Partition nearest neighbor method		A-Bayes partition matching method	
Average positioning error(m)	0.0957	0.0774	Internal area 0.06625	External area 0.1300	Internal area 0.063	External area 0.0821



(a) Distribution map of positioning error



(b) Histogram of positioning error

Fig. 5. Positioning error distribution map and histogram, (a) Distribution map of positioning error, (b) Histogram of positioning error.

40 cm. Only a few test points have a positioning error of more than 25 cm, and most test points have a positioning error of less than 5 cm. It can also be seen from Fig. 5(b) that the error distribution in the middle area is relatively uniform, and the positioning error of the edge area at individual test points is relatively large. In order to further illustrate the positioning effect, Fig. 6(a) shows the Cumulative Distribution Function of the overall positioning error and Fig. 6(b) shows the CDF diagrams of the inner and outer regions.

It can be seen from Fig. 6(a) that the positioning error of the A-Bayes sub-region matching algorithm is 90% of the total test points within 15 cm, and more than 95% within 25 cm. It can be seen from Fig. 6(b) that whether it is the inner area or the outer area, the positioning error with in 15 cm accounts for more than 80% of the test points. The simulation results show that the average positioning error in the center area is 5.62 cm, the average positioning error in the edge area is 2.94 cm, and the overall average positioning error is 4.38 cm. This shows that the positioning effect of the A-bayes sub-region matching algorithm is better, the positioning accuracy is higher, and the positioning speed is significantly faster than other fingerprint matching algorithms.

4. Experimental

In order to further evaluate the performance of the positioning algorithm, this paper conducts a test within the space range $3m \times 3m$, as shown in Fig. 7(a) below. During the experiment, the ground is divided into 361 grids, and at each grid intersection, the device shown in Fig. 7(b) below is used for testing.

In the off-line phase, the output voltage of four PD is measured at each grid intersection point by oscilloscope, and then the received optical power is calculated to establish fingerprint database. In the online positioning stage, some positions of the central region and the edge area are randomly selected as the test points, and the distance between each test point and the grid intersection point is 2 cm. Fig. 8 shows the physical picture of the receiver. There are four hamatsu S1133-14 silicon photodiodes on the receiver. The output current of each silicon photodiode is amplified by TLC2201 trans impedance amplifier.

When some positions of the central region and the edge region are randomly selected as test points, the positioning error distribution histogram and CDF diagram of the center and edge area are shown in Fig. 9(a) and (b), respectively.

It can be seen from Fig. 9(a) that in the actual test process, the positioning error of most test points is less than 5 cm, and the positioning error of a few test points is greater than 30 cm. It can be seen from Fig. 9(b) that in the actual test process, the test points with positioning error less than 15 cm account for more than 60% of the total test points, whether in the central area or the edge area. The experimental results show that the average positioning error of the center area is 9.57 cm, the average positioning error of the edge area is 8.78 cm, and the overall average positioning error is 9.27 cm. It can be seen that there is a certain gap between the experimental test results and the simulation test results. The specific reasons may be the inevitable environmental noise and system noise in the test process, the interference of human factors in the test process and the limited accuracy of the detector itself, which lead to the inaccurate test results of the received optical power experiment. Later, if we can find more precise measuring instruments and more accurate measurement methods, the positioning accuracy will be further improved.

5. Conclusions

In this paper, a single LED lamp is used to realize indoor high-precision positioning. The receiver is composed of one horizontal PD

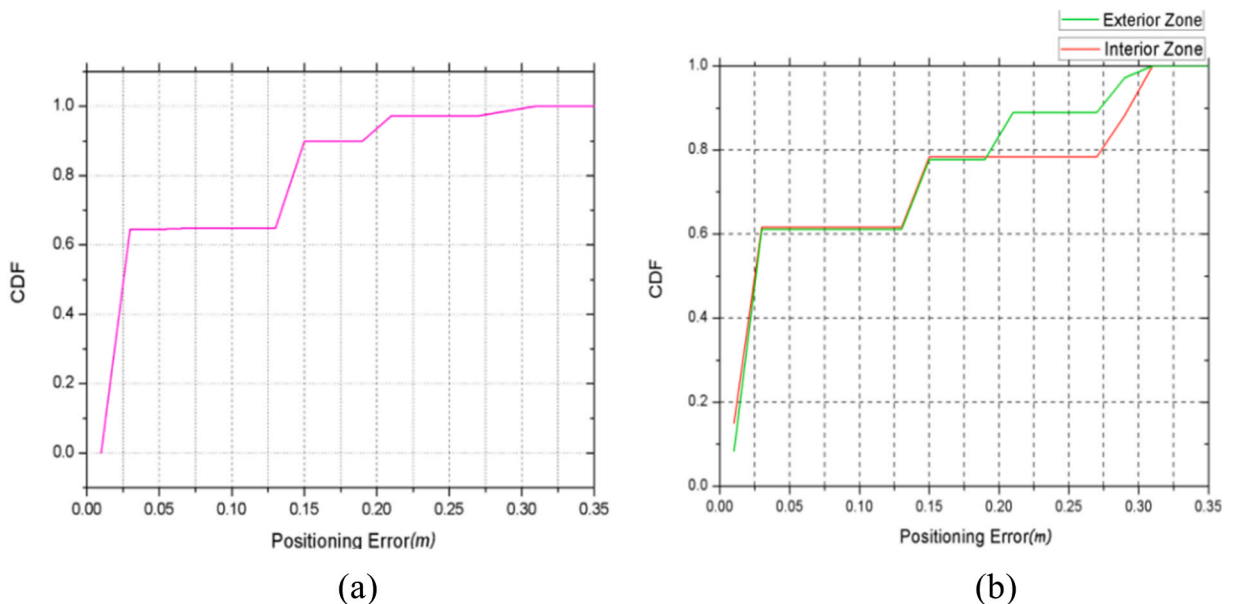


Fig. 6. CDF diagram of positioning error.

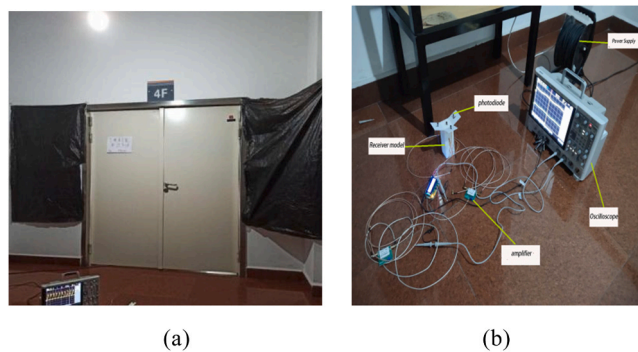


Fig. 7. Positioning experiment environment.

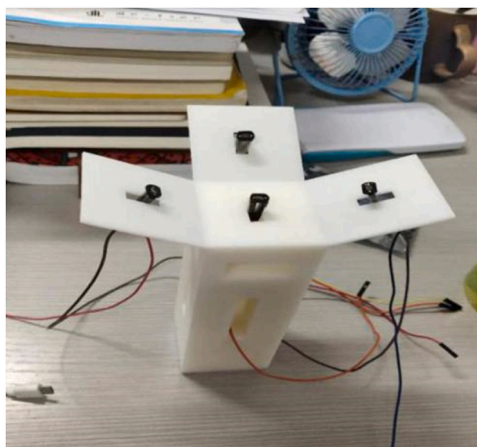


Fig. 8. physical picture of receiver.

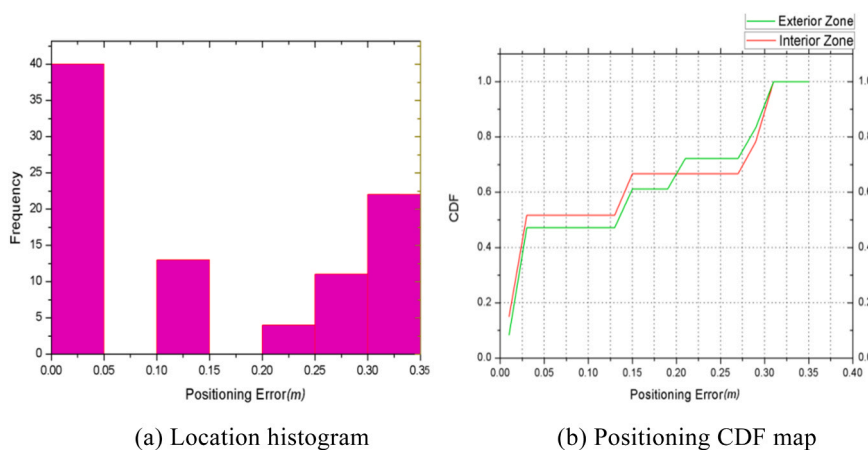


Fig. 9. location error distribution and internal and external CDF diagram, (a) Location histogram, (b) Positioning CDF map.

and three inclined PD. In the off-line phase, under the direct line of sight link and the reflected line of sight link, the received optical power of each PD is measured, and the fingerprint database is established. AdaBoost strong classifier is used to divide the test space into two areas: the center and the edge. In the online positioning stage, for the measured received optical power, the region to which it belongs is first determined, and then the Bayesian algorithm is used to match to complete the positioning. The simulation results show that the average positioning error of the center area is 5.62 cm, the average positioning error of the edge area is 2.94 cm, and the overall average positioning error is 4.38 cm. The experimental results show that the average positioning error of the center area is

9.57 cm, the average positioning error of the edge area is 8.78 cm, and the overall average positioning error is 9.27 cm. For the same test point, the localization algorithm proposed in this paper can significantly shorten the positioning time and has higher positioning accuracy compared with the global nearest neighbor matching method, the regional nearest neighbor matching method and the global Bayesian matching method.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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